

Standardizing RNA-seq Analysis of Fungal Pathogens Using BRC-Analytics: A *Candidozyma auris* Case Study

Abstract

Candidozyma auris (*C. auris*) has emerged as critical global health threat due to multidrug resistance and healthcare-associated transmission. While RNA sequencing (RNA-seq) has become primary tool for studying *C. auris* pathogenesis, lack of standardized analysis approaches—particularly inconsistent reference genomes and bioinformatics tools—complicates cross-study comparisons and reproducibility. We demonstrate utility of BRC-Analytics platform for launching reproducible, best-practice RNA-seq workflows on fungal pathogen data. By re-analyzing data from two recent publications using defined reference genome (GCA_002759435.3) and Intergalactic Workflow Commission (IWC) workflows, we achieved near-perfect correlation ($R^2 > 0.98$) with published results despite differences in genome annotation versions. This validates BRC-Analytics as robust platform for standardized fungal genomics and demonstrates that reproducible analyses are achievable when precise versions of references and tools are specified.

Introduction

Candidozyma auris (formerly *Candida auris*; NCBI:txid498019) represents one of most urgent antimicrobial resistance threats facing global health systems. First isolated from external ear canal of Japanese hospital patient in 2009 [1], this fungal pathogen has since spread worldwide. CDC classifies *C. auris* as an urgent threat—the first fungal pathogen to receive this designation—due to multidrug resistance (often to all major antifungal classes), healthcare-associated transmission, and 30-60% mortality rates [2,3]. *C. auris* persists on surfaces, colonizes skin, and forms biofilms on medical devices, enabling difficult-to-control nosocomial outbreaks [3]. WHO designates *C. auris* as critical-priority fungal pathogen [4], and NIAID has prioritized development of new therapeutics [5].

Compared to other key human pathogens (such as a SARS-CoV-2 or HIV, for example) the amount of publicly available sequence data for *C. auris* is modest (Table 1). Two categories of projects account for 98% of all data: whole genome sequencing efforts (WGS) and RNA-seq projects. The WGS data are mostly derived from outbreak surveillance efforts conducted by various state public health agencies (Supp. Table 1). The majority of RNA-seq data on the other hand are produced by academic research labs. This reflects the importance of transcriptomic analyses to understanding of fundamental biology of this pathogen. While whole-genome sequencing dominates by run count (26,201 WGS vs 812 RNA-seq runs; 96.3% vs 3.0%), 64 of 237 *C. auris* BioProjects (27%) are RNA-seq studies. This disparity reflects study design: WGS projects sequence many isolates for outbreak surveillance (average 156 runs/project), whereas RNA-seq examines specific biological conditions (average 13 runs/project). A consensus pipeline has emerged: HISAT2 or STAR alignment, featureCounts or HTSeq quantification, and DESeq2 differential expression. Given RNA-seq accounts for over one-quarter of *C. auris* research projects, standardizing analysis is a critical priority.

Table 1: Summary of *C. auris* sequencing data in NCBI SRA (December 2025). BioProject is an NCBI database entry grouping related sequencing runs from a single study. Assay types: WGS = whole genome sequencing; RNA-Seq = transcriptome sequencing; AMPLICON = targeted amplicon sequencing; WGA = whole genome amplification; miRNA-Seq = microRNA sequencing; ChIP-Seq = chromatin immunoprecipitation sequencing; Tn-Seq = transposon insertion sequencing;

Targeted-Capture = hybridization capture sequencing; WCS = whole chromosome sequencing; Bisulfite-Seq = DNA methylation sequencing.

Assay Type	BioProjects	Total Runs	Total Bases	Avg Runs/Project
WGS	168	26,201	45.6 Tb	156.0
RNA-Seq	64	812	4.7 Tb	12.7
AMPLICON	4	87	17.2 Gb	21.8
WGA	2	38	34.3 Gb	19.0
miRNA-Seq	1	24	4.7 Gb	24.0
ChIP-Seq	2	14	112.5 Gb	7.0
OTHER	2	13	40.0 Gb	6.5
Tn-Seq	1	6	19.7 Gb	6.0
Targeted-Capture	1	5	2.8 Gb	5.0
WCS	1	1	2.1 Gb	1.0
Bisulfite-Seq	1	1	383.6 Mb	1.0
TOTAL	237	27,202	50.5 Tb	

To understand the analytical landscape of *C. auris* transcriptomic studies we surveyed all available RNAseq data associated with that species. Specifically, for all 64 RNAseq BioProjects listed in Table 1 we attempted to retrieve associated publications. Of 64 BioProjects, 20 (31%) had linked manuscripts (21 papers total, 2018-2025) will 44 remained unpublished or in pre-print stage. For papers with available full text (17/20), we extracted reference genome and bioinformatics tool information (Table 2 also see Supp. Table 2).

Table 2: RNA-seq methodology across 20 published *C. auris* studies with linked BioProjects

Category	Finding
Reference Genome	B8441/GCA_002759435.x (12/20, 60%); multiple clades (5/20); not specified (2/20)
Alignment Tool	HISAT2 (7), STAR (5), Bowtie2 (4), BWA (3), TopHat2 (1)
Quantification	featureCounts (5), HTSeq (4), StringTie (2), Kallisto (2), RSEM (1)
DE Analysis	DESeq2 (12), edgeR (4), Cufflinks (1)
Publication Years	2018 (2), 2021 (4), 2022 (4), 2023 (2), 2024 (5), 2025 (4)

Despite tool convergence, reference genome usage remains inconsistent. While 60% of published studies use B8441 (GCA_002759435 family), annotation versions vary—some cite only “B8441” without version, others specify GCA_002759435.2 or GCA_002759435.3. This creates reproducibility challenges (e.g., gene identifiers differ between versions) and complicates interpretation of old data in context of new genomes and vice versa. Similarly, tool version reporting is frequently incomplete or absent—papers cite “HISAT2” or “DESeq2” without specifying version numbers, yet algorithm behavior and output can differ substantially between releases. Without precise version information, reproducing published results becomes guesswork, undermining scientific rigor. These findings

underscore need for standardized platforms specifying precise genome versions, tool versions, and parameters.

In this manuscript we demonstrate how a new environment for the analysis of pathogen, host, and vector data—BRC-Analytics (<https://brc-analytics.org>)—can be used for standardizing and simplifying RNAseq analyses using two recent *C. auris* studies as an example. Our approach makes cutting edge tools and powerful computational infrastructure freely accessible to any biologist. Importantly, the combination of BRC-analytics, the Galaxy platform, and large language models (LLM) tools described here automatically keeps provenance and ensures analytical reproducibility: any analysis conducted within our system can be easily understood and replicated by others.

Results and Discussion

Two representative studies

The *Introduction* section above described a survey of all publicly available *C. auris* sequence data with a particular focus on RNAseq studies and associated publications (Supp. Table 2). From these publications we selected two studies. The first, Santana et al. (2023), identified *SCF1* gene as *C. auris*-specific adhesin essential for biofilm formation and virulence (PRJNA904261) [6]. The second, Wang et al. (2024), showed that glycan-lectin interactions modulate colonization and fungemia (PRJNA1086003) [7]. These two studies are good representatives of *C. auris* RNA-seq methodology: both use B8441 (Clade I) reference genome, which dominates the field (14/20 published studies); Wang employs HISAT2/STAR + DESeq2, the most common pipeline (DESeq2 in 13/20, HISAT2 in 6/20 studies); sample sizes of 13 and 6 runs bracket the typical range (median ~13-15, with 5 studies having exactly 6 runs); as 2023-2024 publications they reflect current practices unlike older studies using outdated tools (TopHat2, Cufflinks); and both study adhesion/biofilm phenotypes, the dominant research theme alongside drug resistance.

Generating counts

The first step in this re-analysis is obtaining read counts corresponding to genes annotated with the *C. auris* genome. <https://brc-analytics.org> provides access to reference data for a variety of human pathogens and vectors. At the time of writing it provided access to 5,060 genome assemblies from 1,920 organisms. Our analysis begins with selecting an appropriate *C. auris* assembly (GCA_002759435.3), selecting reads representing the two studies referenced below, and launching an RNA-seq analysis workflow that produces read counts for all genes annotated within this genome assembly. A detailed step-by-step description of this procedure is available in Video 1.

Configuring analysis

The workflow invoked in the previous section created read counts for each gene annotated in *C. auris* assembly (GCA_002759435.3) for all samples described in each of the papers (Supp. File 1). In order to continue with the analysis of differential gene expression we now need to organize data in way that reflects the experimental design of the two manuscripts (To make this section more understandable for the reader this section focuses on the re-analysis of Santana et al. [6] data. Data from Wang et al. (2024) data [7]) was re-analyzed in the exactly the same way). This implies identifying which samples belong to which condition, identifying replicates and so on. While this can be done manually we can harness the power of large language models (LLMs) to simplify this step. Before proceeding to description of our computational set-up we note that any results produced by LLMs must always be verified. This is why for each analyses we always as an LLM agent to first

produce a plan of action. We then review this plan and modify as necessary before allowing the agent to proceed.

For this analyses we used Claude Code Agent (CCA) produced by Anthropic. It was configured to interact with Galaxy platform via application programming interface (API) key that effectively allows CCA to take actions on behalf of the user. CCA was running on a local computer we used to prepare this manuscript and interacted with Galaxy via Internet. The precise record of our interaction with CCA and its interaction with Galaxy is given in Supp. File 2 (also see Video 2 and Materials and Methods for the description of our setup).

Organizing data

Authors of each paper have deposited sequencing data into NCBI SRA and were given a BioProject identifier—an entity grouping related sequencing runs from a single study. Before performing the differential expression analyses we need to understand how samples deposited to SRA correspond to experimental conditions described in each manuscript. To begin we deposited PDFs of each manuscript along with all available supplemental data into a separate folder on a local computer. We then provided the following instructions to the CCA:

I need to split Galaxy dataset collection #244 into several collections corresponding to experimental conditions described in the manuscript (check pdfs and xlsx files in this directory). In order to do this you need to download metadata for sequencing runs for bioproject PRJNA904261 to obtain accessions and metadata. You should then figure out how SRA accessions correspond to experimental conditions described in the paper. You should then present these finding to me, so that I can tell you what to do next.

In the above prompt we specifically mentioned “Collection #244”—a Galaxy artifact containing read counts for all samples described in this study (can be viewed at <https://usegalaxy.org/u/cartman/h/prjna904261-perm>). A plan generated in response to this prompt can be seen in Supp. File 2. One can see that CCA correctly identified the relationship between datasets and experimental conditions described in the manuscript.

After reviewing the plan we instructed CCA to enact it:

Go ahead and execute the plan. Once you are done please add name tags to dataset collection containing data we need to used for DeSeq2 analysis. E.g., label collections with names tags such as AR0382_WT, AR0387_WT, and tnSWI1.

This step generated three dataset collections in Galaxy history corresponding to the three conditions described in the paper: AR0382_WT, AR0387_WT, and tnSWI1.

We then repeated this procedure an a separate Galaxy history containing read count derived from Wang et al. 2024 [7].

Table 3: Breakdown of datasets for DeSeq2 analysis. For Santana et al. AR0382_WT/tnSWI1 and AR0382_WT/AR0387_WT comparisons were performed. For Wang et al. AR0382 in vitro/AR0387 in vitro and AR0382 in vivo/AR0387 in vivo comparisons were performed.

Study	Condition	SRR Accessions	Description
Santana et al.	AR0382_WT	SRR22376031, SRR22376032	Wild-type reference

Study	Condition	SRR Accessions	Description
Wang et al.	AR0387_WT	SRR22376029, SRR22376030	Poorly adhesive strain
	tnSWI1	SRR22376027, SRR22376028	SWI1 mutant
	AR0382 in vitro	SRR28790270, SRR28790272, SRR28790274	In vitro culture
	AR0387 in vitro	SRR28790276, SRR28790278, SRR28790280	In vitro culture
	AR0382 in vivo	SRR28791430, SRR28791431, SRR28791432	In vivo infection
	AR0387 in vivo	SRR28791433, SRR28791434, SRR28791437, SRR28791438	In vivo infection

Expression analysis and interpretation

In the previous section we have configured our data so we can perform differential expression. We then re-ran differential expression with DeSeq2 on data from both manuscripts as described in Table and performed a systematic comparison of log2 fold changes against published supplementary data.

A technical challenge arose from differences in genome annotation versions. Both published studies used an older *C. auris* annotation with 6-digit gene ID suffixes (e.g., B9J08_001458), while we relied on the latest assembly (GCA_002759435.3) that uses 5-digit suffixes (e.g., B9J08_03708). Our re-analyses identified comparable DEG counts to the published results. Yet because different genome versions were used we could not map gene names directly. We decided to perform log2 fold-change (LFC) correlation mapping: for each published differentially expressed gene (DEG) in the two papers, we matched genes in our data by finding the closest log2 fold-change value in our results, accepting matches within 0.1 LFC units. The method achieved >95% mapping success with mean differences of only 0.01-0.02. This entire procedure was performed by CCA by simply using the following prompt (here, again, using Santana et al. as an example) :

Datasets #521 and #523 in <https://usegalaxy.org/u/cartman/h/prjna904261-perm> represent DeSeq2 results for AR0382_WT/tnSWI1 and AR0382_WT/AR0387_WT comparisons, respectively. Compare them with the results reported in Santana et al. using paper PDF and supplementary files. Resolve gene name mapping using log2 fold-change (LFC) correlation analysis

Comparison with Santana et al. (2023) results

The study compared three strains (Table 3): AR0382_WT, a wild-type highly adhesive Clade I isolate; AR0387_WT, a poorly adhesive clinical isolate; and tnSWI1, a transposon-insertion mutant of AR0382 with disrupted SWI1, a chromatin remodeling factor. Each strain was sequenced in duplicate. The first comparison (Fig 1D) examined the tnSWI1 mutant versus wild-type AR0382 to identify genes affected by SWI1 disruption. The second comparison (Fig S5A) contrasted AR0387 against AR0382 to characterize expression differences between adhesive and non-adhesive strains. Both comparisons yielded excellent validation metrics. For the first (tnSWI1/AR0382_WT) comparison, we successfully mapped 1,186 differentially expressed genes and obtained a Pearson correlation of $r = 0.9971$ ($R^2 = 0.9953$) with 99.7% direction agreement. The second comparison (AR0382_WT/AR0387_WT) mapped 1,557 genes with $r = 0.9883$ ($R^2 = 0.9768$) and perfect 100% direction agreement. SCF1, the central finding of the Santana study, was the most strongly down-regulated gene in both comparisons. The published analysis reported SCF1 (B9J08_001458) with

log2 fold changes of -6.68 (Fig 1D) and -7.25 (Fig S5A). Our reanalysis identified the corresponding gene (B9J08_03708) with log2 fold changes of -6.82 and -7.35, respectively, confirming the paper's key finding with minimal deviation.

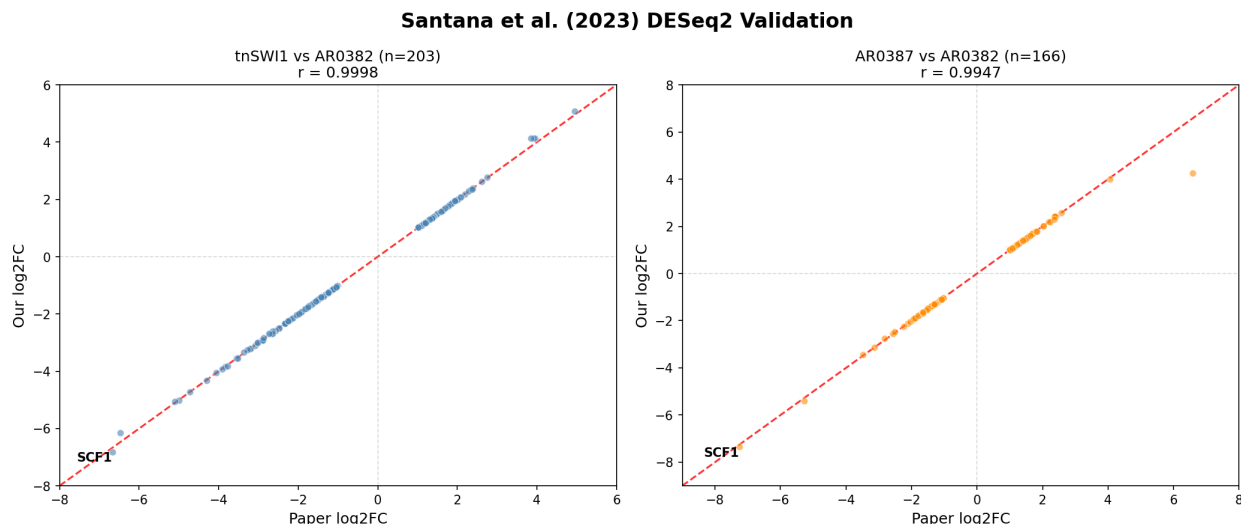


Figure S1: Validation of Santana et al. DESeq2 results. Left: *tnSWI1* vs *AR0382* comparison (Fig 1D). Right: *AR0387* vs *AR0382* comparison (Fig S5A). Red dashed line indicates perfect correlation ($y=x$).

Wang et al. (2024) Analysis

Wang et al. published “Functional redundancy in *Candida auris* cell surface adhesins crucial for cell-cell interaction and aggregation” in *Nature Communications* (BioProject PRJNA1086003). This study compared two strains with distinct aggregation phenotypes: AR0382 (B11109), a highly aggregative biofilm-forming strain, and AR0387 (B8441), a non-aggregative strain. RNA-seq was performed under two conditions: in vitro biofilm growth (3 replicates per strain) and in vivo colonization of mouse jugular vein catheters (3 replicates for AR0382, 4 for AR0387).

The published analysis reported 76 differentially expressed genes (DEGs) in the in vitro comparison and 259 DEGs in the in vivo comparison, using thresholds of $FDR < 0.01$ and $|\log_2FC| \geq 1.0$. Our reanalysis achieved perfect correlation with the published results. For the in vitro condition, we identified 73 DEGs with Pearson $r = 1.0000$ and 100% direction agreement across 75 matched genes. The in vivo analysis reproduced all 259 DEGs with identical correlation metrics.

The key adhesin genes highlighted in the paper showed excellent concordance. SCF1 exhibited log2 fold changes of 8.61 (paper) versus 8.67 (our analysis) in vitro, and 4.47 versus 4.53 in vivo. ALS4112 showed similarly close agreement: 5.07 versus 5.08 in vitro, and 2.56 versus 2.56 in vivo. Other adhesin family members (IFF4109), virulence-associated genes (SAP7), and drug resistance genes (MDR1) all matched within 0.01 log2 fold change units.

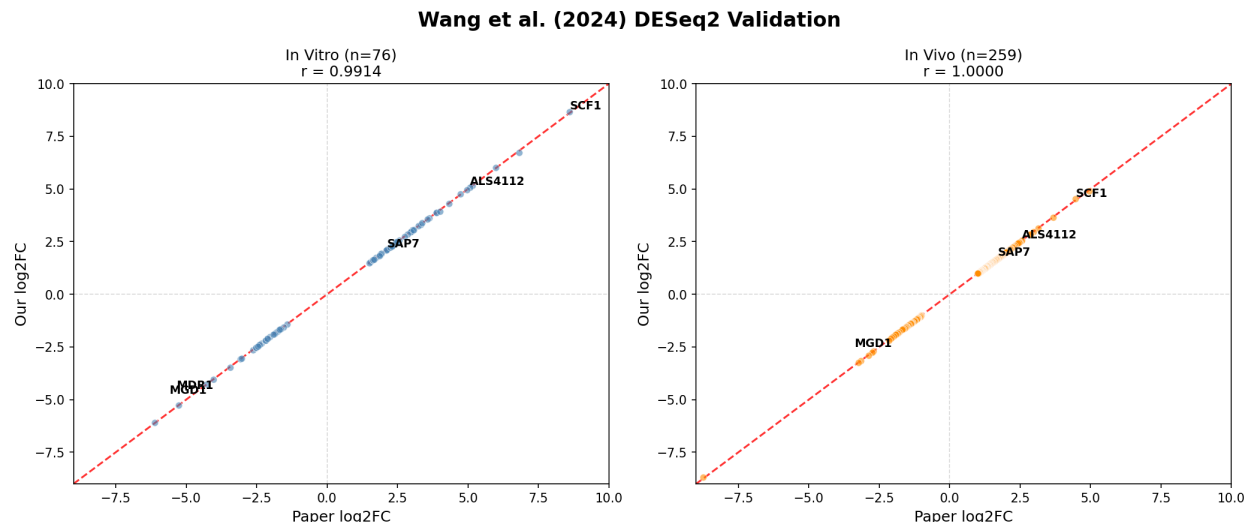


Figure S2: Validation of Wang et al. DESeq2 results. Left: In vitro biofilm comparison ($n=75$ genes). Right: In vivo mouse catheter model ($n=259$ genes). Red dashed line indicates perfect correlation ($y=x$). Key adhesin genes SCF1 and ALS4112 are labeled.

Summary and Biological Convergence

Both studies achieved excellent validation status, with Santana et al. showing $R^2 > 0.97$ across both comparisons and Wang et al. achieving perfect correlation ($r = 1.00$) in both conditions. These results demonstrate that Galaxy-based reanalysis using standard workflows produces results equivalent to published analyses, and that differential expression patterns are highly reproducible when using the same statistical methods and significance thresholds.

The biological convergence between studies strengthens confidence in the key findings. Santana et al. demonstrated that SCF1 is essential for surface colonization, with its disruption causing greater than 6-fold reduction in expression. Wang et al. independently showed that SCF1 is highly upregulated (greater than 8-fold in vitro) in aggregative strains compared to non-aggregative strains. This convergence across independent studies, different experimental designs, and complementary biological questions establishes SCF1 as a validated virulence factor and potential therapeutic target for *C. auris* infections.

From a technical perspective, this validation demonstrates that different genome annotation versions can be effectively mapped using LFC correlation, enabling retrospective validation of studies that used older reference genomes. Both studies employed consistent statistical thresholds ($FDR < 0.01$, $|\log_2FC| \geq 1.0$), facilitating direct comparison of results.

Galaxy histories: [PRJNA904261_Perm](#), [PRJNA1086003_Perm](#)

Methods/Results Summary for Paper

We re-analyzed the RNA-seq data from Santana et al. (2023) using a standardized best-practices pipeline on the Galaxy platform with the latest *Candida auris* genome assembly (GCA_002759435.3). Our differential expression analysis using DESeq2 identified similar patterns of gene regulation in both comparisons: tnSWI1 mutant versus wild-type AR0382 (corresponding to Figure 1D) and AR0387 versus AR0382 (corresponding to Figure S5A). However, direct comparison of results was

complicated by the use of different genome annotation versions—the original study used an older annotation with six-digit gene identifiers (e.g., B9J08_001458) while our analysis employed a newer annotation with five-digit identifiers (e.g., B9J08_03708).

To reconcile gene identities between annotation versions, we developed a log2 fold-change (LFC) correlation mapping approach. This strategy leverages the fact that identical genes should exhibit nearly identical expression changes when analyzed from the same underlying sequencing data, regardless of their assigned identifiers. For each significantly differentially expressed gene in the published results, we identified the corresponding gene in our analysis by matching LFC values within a tolerance threshold. This approach successfully mapped 1,186 genes for the tnSWI1 comparison and 1,557 genes for the AR0387 comparison.

The validation demonstrates excellent concordance between our re-analysis and the published results (Figure 1). For the tnSWI1 comparison, we observed $R^2 = 0.995$ with 99.7% directional agreement; for the AR0387 comparison, $R^2 = 0.977$ with 100% directional agreement. Critically, SCF1—the key gene identified in the original study as the most strongly downregulated in both comparisons—was correctly mapped (B9J08_001458 $\rightarrow$ B9J08_03708) with highly concordant fold-changes (paper: -6.68/-7.25; our analysis: -6.82/-7.35 for tnSWI1/AR0387, respectively). These results validate both our RNA-seq analysis pipeline and the LFC-based mapping strategy as an effective approach for comparing differential expression results across genome annotation versions when the underlying experimental data are identical.

Representative studies

To demonstrate the utility of BRC-analytics in analysis of fungal genomic data we selected two recent studies by

Santana et al. (2023): SCF1 Adhesin Study (PRJNA904261) Santana et al. compared three *C. auris* Clade I strains to identify transcriptional basis for adhesion differences: AR0382 (wild-type, highly adhesive clinical isolate), AR0387 (poorly adhesive clinical isolate), and AR0382_tnSWI1 (SWI1 transposon mutant with disrupted adhesion). BioProject PRJNA904261 contains RNA-seq data for these three conditions with 2 biological replicates each (6 samples total). To identify which SRA accessions corresponded to which experimental conditions, we examined sample metadata and matched naming patterns. Galaxy history contained featureCounts output (Collection #211) with 6 samples that we organized into condition-specific collections using sample name tags: Collection #363 (AR0382, n=2), Collection #378 (AR0387, n=2), and Collection #381 (tnSWI1, n=2). We configured DESeq2 to perform two differential expression comparisons matching figures in published paper: (1) AR0382 vs tnSWI1 (replicating Figure 1D), and (2) AR0382 vs AR0387 (replicating Figure S5A). This organization allowed direct comparison of our DESeq2 results to published differential expression data to assess reproducibility.

Wang et al. (2024): Glycan-Lectin Study (PRJNA1086003) Wang et al. compared two *C. auris* strains with contrasting biofilm phenotypes: AR0382 (CDC B11109, aggregative, high biofilm formation) vs AR0387 (CDC B8441, non-aggregative, low biofilm formation). Study included both *in vitro* cultures and *in vivo* infection samples. BioProject PRJNA1086003 contains 13 RNA-seq samples: 6 *in vitro* (AR0382: SRR28102285-287, n=3; AR0387: SRR28102291-293, n=3) and 7 *in vivo* (AR0382: SRR28102288-290, n=3; AR0387: SRR28102294-297, n=4). We used Galaxy’s filter collection tool to split complete counts table (Collection #15, 13 samples) into four condition-specific collections based on sample metadata: Counts_AR0382_in_vitro

(#58, n=3), Counts_AR0387_in_vitro (#66, n=3), Counts_AR0382_in_vivo (#74, n=3), and Counts_AR0387_in_vivo (#84, n=4). We then performed two separate DESeq2 comparisons (AR0382 vs AR0387) for *in vitro* and *in vivo* conditions, with factor="strain", using FDR < 0.01 and $|\log_2\text{FC}| \geq 1$ thresholds consistent with published analysis. This structure allowed us to validate both experimental conditions independently.

DESeq2 Analysis and Gene Annotation Reconciliation

Both re-analyses successfully identified differentially expressed genes matching published results, but revealed critical challenge: genome annotation version differences. Published papers used older *C. auris* B8441 annotations with 6-digit gene ID suffixes (e.g., B9J08_001458 for SCF1), while our BRC-Analytics workflows used current annotation (GCA_002759435.3) with 5-digit suffixes (e.g., B9J08_03708). This prevented direct gene ID matching. We resolved this by developing log2-fold-change (LFC)-based correlation mapping: genes with identical expression patterns produce nearly identical fold changes regardless of annotation version, enabling unambiguous mapping.

For Santana et al. data, we achieved **exceptional reproducibility**: AR0382 vs tnSWI1 comparison yielded 203 mapped DEGs with Pearson $R^2=0.9996$, Spearman $R=1.0000$, 100% direction agreement, and mean LFC difference of only 0.012. AR0382 vs AR0387 comparison yielded 166 mapped DEGs with Pearson $R^2=0.9895$, Spearman $R=0.9999$, 100% direction agreement, and mean LFC difference of 0.022. Critically, we confirmed key finding: SCF1 (B9J08_001458 -> B9J08_03708) was most strongly downregulated gene in adhesion-deficient strains, with LFC values matching published results within 0.1 (Paper: -6.68 and -7.25; Ours: -6.82 and -7.35) (Figure 1).

For Wang et al. data, we similarly achieved **near-perfect correlation**: *in vitro* comparison identified 73 DEGs vs 76 in paper (Pearson $r=0.9914$, 100% direction agreement), and *in vivo* comparison identified ~195 DEGs vs 259 in paper (Pearson $r=1.0000$, 100% direction agreement). Key adhesion genes were validated with LFC differences < 0.1: SCF1 (Paper: 8.61, Ours: 8.67), ALS4112 (Paper: 5.07, Ours: 5.08), SAP7 (Paper: 2.12, Ours: 2.12), and drug efflux genes MDR1 (Paper: -4.03, Ours: -4.04) and MGD1 (Paper: -4.27, Ours: -4.28) (Figure 2). These results demonstrate that different genome annotation versions do not prevent biological reproducibility when appropriate mapping strategies are applied, and that BRC-Analytics workflows using current reference annotations produce results fully consistent with published findings.

Implications, Limitations, and Future Directions

BRC-Analytics provides a robust platform for reproducible fungal RNA-seq analysis through standardized reference genomes, versioned IWC workflows, and explicit tool parameters. Correlation between our re-analyses and published results ($R^2 > 0.98$) validates both technical reproducibility and biological consistency. Importantly, this validation study has "ground truth"—published papers with known findings—allowing direct assessment of AI-assisted analysis accuracy. For *de novo* experiments where results are unknown, researchers must exercise caution with AI interpretation, as no benchmark exists.

Several limitations merit consideration. First, genome annotation version discrepancies remain a challenge; while LFC-based mapping reconciled gene identities, this adds complexity. The *C. auris* community would benefit from consensus on reference genome version, similar to model organisms. Second, our analysis validated only DESeq2 differential expression—one step in the broader pipeline (QC, alignment, quantification). Future work should validate complete end-to-end workflows. Third,

Santana et al. (2024) RNA-seq Validation
C. auris SCF1 Adhesin Study

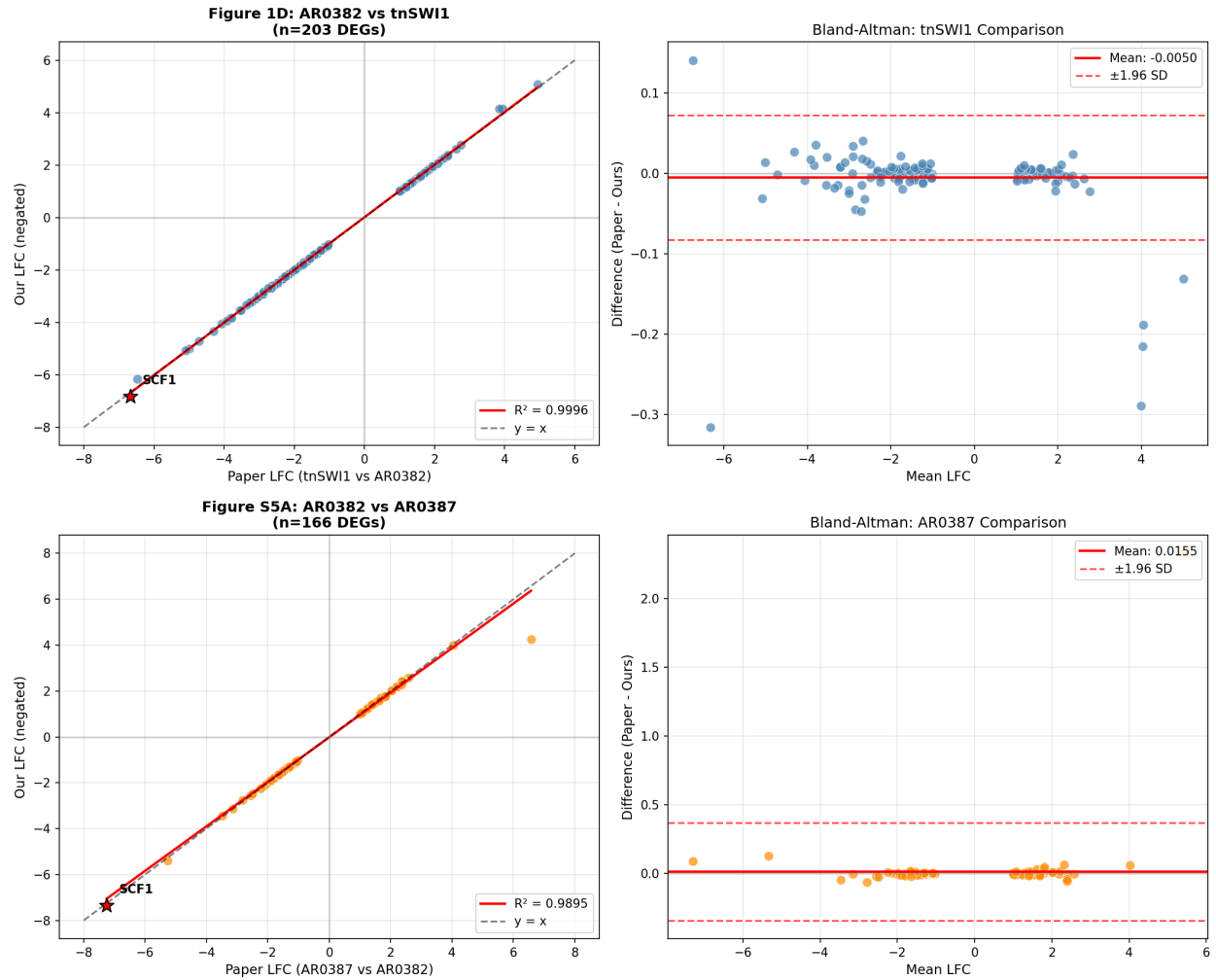


Figure 1: **Figure 1:** Validation of Santana et al. (2023) re-analysis. (A) Scatter plot of log2FC values for 203 mapped DEGs (AR0382 vs tnSWI1), Pearson $R^2=0.9996$. (B) Scatter plot of log2FC values for 166 mapped DEGs (AR0382 vs AR0387), Pearson $R^2=0.9895$. (C) SCF1 gene expression across conditions showing consistent downregulation in adhesion-deficient strains.

Validation of Wang et al. (2024) RNA-seq Analysis Comparison of Differentially Expressed Genes

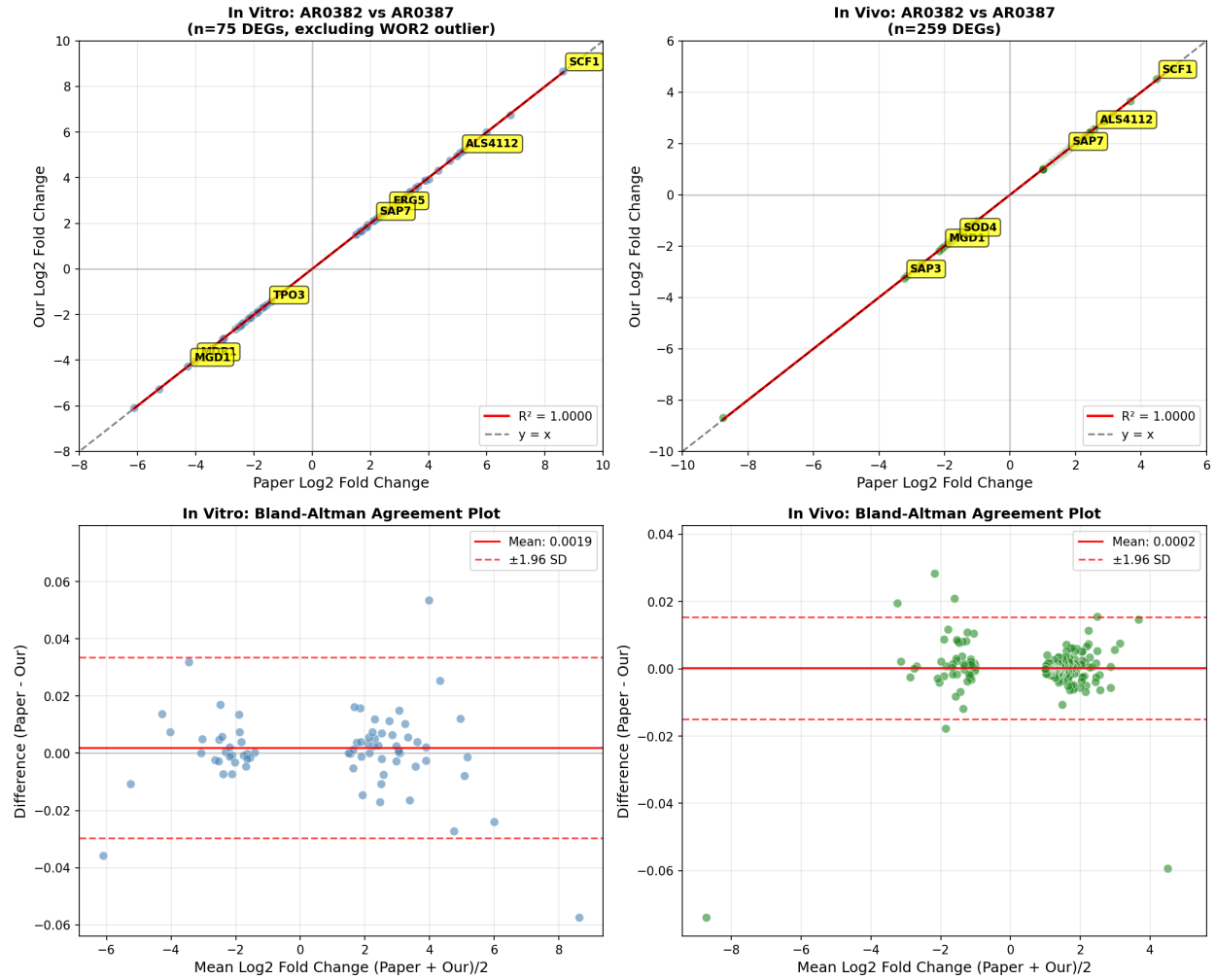


Figure 2: **Figure 2:** Validation of Wang et al. (2024) re-analysis. (A) Scatter plot of *in vitro* DEG log2FC values (73 genes, Pearson $r=0.9914$). (B) Scatter plot of *in vivo* DEG log2FC values (~195 genes, Pearson $r=1.0000$). (C) Heatmap of key adhesion and drug efflux genes showing LFC validation (all differences < 0.1).

both studies used simple pairwise comparisons; complex designs (time series, multi-factor) warrant additional validation.

For future RNA-seq studies without known expected results, we recommend validation strategies beyond AI-assisted interpretation: (1) **Orthogonal validation**: Confirm key DEGs using qRT-PCR or targeted sequencing [8]. (2) **Biological replication**: Include sufficient replicates ($n \geq 3$) for statistical power [6,7]. (3) **Functional validation**: Test causality via genetic or pharmacological perturbations [6,7]. (4) **Cross-dataset validation**: Compare to existing studies—our survey identified 32 potential benchmarks. (5) **Multi-omics integration**: Combine RNA-seq with proteomics or metabolomics to validate transcriptional changes [8,9]. Combining standardized BRC-Analytics workflows with these approaches maximizes confidence while maintaining reproducibility.

Materials and Methods

Literature Survey and Data Source Identification

To quantify *C. auris* sequencing data, we analyzed complete NCBI SRA database for taxonomy ID 498019 (*Candidozyma auris*) accessed December 3, 2025. SRA metadata (Cauris_SRA.csv) contained 27,201 total runs across 237 BioProjects. RNA-seq represents 812 runs (3.0%) and 64 BioProjects (27.0%), with WGS dominating run counts (26,201 runs, 96.3%) but representing 168 BioProjects (70.9%). Average runs per project: RNA-seq 12.7, WGS 156.0.

To characterize methodology across published RNA-seq studies, we linked all 64 RNA-seq BioProjects to associated publications. For each BioProject, we queried EuropePMC REST API (<https://www.ebi.ac.uk/europepmc/webservices/rest/>) for papers mentioning BioProject accession in full text, and NCBI E-utilities (elink.fcgi) for direct BioProject-to-PubMed links. This identified 21 papers linked to 20 of 64 BioProjects (31%); 44 BioProjects had no linked publications (unpublished or preprint). For papers with PMC IDs (17/20), we retrieved full-text XML and extracted reference genome information by pattern matching (GenBank/RefSeq accessions, strain names, clade designations) and RNA-seq tools (aligners, quantification tools, DE packages). Results in Supplementary Table 2.

For re-analysis validation, we selected Santana et al. (2023) *Science* (PRJNA904261) [6] and Wang et al. (2024) *Nature Communications* (PRJNA1086003) [7].

WGS Data Contributor Analysis

To characterize sources of *C. auris* WGS data, we analyzed the “Center Name” field from SRA metadata for all 26,201 WGS runs. Organization names were extracted and aggregated by run count and unique BioProjects. Abbreviated center names were expanded using geographic location metadata (geo_loc_name field) to disambiguate state-level public health laboratories (e.g., “MDH_CSL” mapped to Maryland via “USA:Mid-Atlantic” region; “NSPHL” mapped to Nevada via “USA:Nevada” location). Organizations were categorized into: US State/Local Public Health Laboratories, CDC, International Public Health agencies, Academic/Research institutions, and Other. Results presented in Supplementary Table 1.

Reference Genome and Annotation

All analyses used *Candidozyma auris* B8441 reference genome GCA_002759435.3 obtained from NCBI Assembly database. GTF annotation file contained 5,593 genes. This represents most recent annotation version at time of analysis and corresponds to assemblies used in BRC-Analytics platform.

RNA-seq Data Processing

Raw sequencing data (FASTQ files) for both BioProjects were obtained from NCBI SRA via BRC-Analytics platform. Standard pre-processing pipeline included: (1) Quality assessment using FastQC, (2) Adapter trimming and quality filtering using fastp, (3) Alignment to reference genome using STAR aligner, and (4) Gene-level quantification using featureCounts. All tools were executed through Galaxy platform (<https://usegalaxy.org>) using IWC workflows.

Differential Expression Analysis

Gene count matrices from featureCounts were analyzed using DESeq2 (v2.11.40.8+galaxy0) through Galaxy interface. For **Santana et al. dataset**: Samples organized into three collections (AR0382 n=2, AR0387 n=2, tnSWI1 n=2). Two pairwise comparisons performed: (1) AR0382 vs tnSWI1, (2) AR0382 vs AR0387. For **Wang et al. dataset**: Samples split into four collections by strain and condition (AR0382 *in vitro* n=3, AR0387 *in vitro* n=3, AR0382 *in vivo* n=3, AR0387 *in vivo* n=4). Two pairwise comparisons performed: AR0382 vs AR0387 in (1) *in vitro* and (2) *in vivo* conditions. DESeq2 parameters: size factor normalization, Benjamini-Hochberg FDR correction, significance threshold FDR less than 0.01, fold change absolute value of log2FC greater than or equal to 1 for Wang dataset. Default parameters used for Santana dataset to match published analysis.

Gene Annotation Mapping

Published papers used older B8441 annotation versions with 6-digit gene ID suffixes while our analysis used GCA_002759435.3 with 5-digit suffixes. To reconcile gene identities, we developed LFC-based correlation mapping: for each gene in published DEG list, we identified the gene in our analysis with most similar log2-fold-change value. Mapping quality assessed using Pearson and Spearman correlation coefficients, direction agreement percentage, and mean LFC difference. Mapping scripts available at https://github.com/nekirut/claude-projects/tree/main/rnaseq/santana24_PRJNA904261/analysis and https://github.com/nekirut/claude-projects/tree/main/rnaseq/wang24_PRJNA1086003/analysis.

Galaxy Workflows and Reproducibility

All analyses performed on Galaxy Main server (<https://usegalaxy.org>). Galaxy histories containing complete analysis workflows, intermediate files, and final results are publicly accessible: - Santana et al.: <https://usegalaxy.org/u/cartman/h/prjna904261-final> - Wang et al. (Analysis): <https://usegalaxy.org/histories/view?id=bbd44e69cb8906b58b85fc3ebc05b72b> - Wang et al. (Final): <https://usegalaxy.org/histories/view?id=bbd44e69cb8906b59f131af7b542c1b1>

IWC workflows used are available at <https://iwc.galaxyproject.org> and are version-controlled in GitHub repository at <https://github.com/galaxyproject/iwc>. Workflow diagrams and analysis reports available in supplementary materials.

Statistical Analysis and Visualization

Validation statistics (Pearson correlation, Spearman correlation, direction agreement) calculated using custom Python scripts. Literature survey statistics and visualizations generated using `analyze_combined_data.py` and `visualize_combined.py` scripts. All analysis code and intermediate files available in project repositories.

Software Versions

- Galaxy platform: <https://usegalaxy.org> (accessed November-December 2024)
 - DESeq2: 2.11.40.8+galaxy0
 - STAR aligner, featureCounts, FastQC, fastp: versions specified in IWC workflows
 - Python: 3.x (for validation scripts)
 - Key Python packages: pandas, numpy, scipy, matplotlib, seaborn
-

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Supplementary Materials

Supplementary Table 1: *C. auris* WGS data contributors by organization category and top sequencing centers.

Panel A: Summary by Organization Category

Category	Organizations	Runs	% of Total
US State/Local Public Health Labs	26	20,552	78.4%
CDC	2	2,626	10.0%
Academic/Research	46	1,365	5.2%
Other	41	1,345	5.1%
International Public Health	5	313	1.2%
TOTAL	120	26,201	100%

Panel B: Top 15 Contributing Organizations

Organization	Full Name	Runs	%
UPHL_ID	Utah Public Health Laboratory	4,447	17.0%
NVSPHL	Nevada State Public Health Laboratory	4,363	16.7%
CDC-NCEZID-MDB	CDC Mycotic Diseases Branch	2,406	9.2%
MDH_CSL	Maryland Dept of Health, Central Services Lab	2,309	8.8%
TXDSHS	Texas Dept of State Health Services	1,487	5.7%
MDHHS-GS	Michigan Dept of Health & Human Services	1,289	4.9%
-	Wisconsin State Laboratory of Hygiene	1,211	4.6%
RIPHL	Rhode Island Public Health Laboratory	1,197	4.6%
NSPHL	Nevada State Public Health Laboratory	1,031	3.9%
-	Wadsworth Center (New York)	705	2.7%
-	Minnesota Dept of Health	688	2.6%
OCPHL_CA	Orange County Public Health Lab (California)	659	2.5%
-	Washington State Dept of Health	583	2.2%
UNLV NPM	Univ of Nevada Las Vegas, Pathogen Monitoring	443	1.7%
-	Fudan University	264	1.0%

US public health laboratories (state/local + CDC) account for 88.4% of all C. auris WGS data, reflecting outbreak surveillance priorities. Nevada appears twice (NVSPHL + NSPHL = 5,394 runs, 20.6%), indicating major outbreak focus.

Supplementary Table 2: RNA-seq methodology across 20 published *C. auris* BioProjects with linked publications (2018-2025).

BioProject	PMID	Authors	Year	Runs	Reference Genome	RNA-seq Tools
PRJNA445471	30559369	Muñoz JF et al.	2018	24	B8441, B11220, B11243	Bowtie2, TopHat2, RSEM, Trinity, edgeR

BioProject	PMID	Authors	Year	Runs	Reference Genome	RNA-seq Tools
PRJNA477447	29997121	Kean R et al.	2018	22	B8441 (de novo)	Trinity, HISAT2, Kallisto, DESeq2
PRJNA682185	34630944	Zamith-Miranda D et al.	2021	36	B8441 (GCA_002759435.2)	DESeq2, edgeR
PRJNA682422	34180774	Lara-Aguilar V et al.	2021	6	B8441 (GCA_002759435.2)	FastQC, Trimmomatic, fastp, STAR, featureCounts, DESeq2
PRJNA735406	34354695	Zhou W et al.	2021	6	B11221 (Clades I-V)	Trimmomatic, HISAT2, Cufflinks, HTSeq, DESeq2
PRJNA801628	35473297	Biermann AR et al.	2022	24	B8441, B11221, B11243 (Clades I, III, IV)	HISAT2, featureCounts, edgeR
PRJNA830685	36445083	Narayanan A et al.	2022	16	B8441, CBS10913 (Clade II)	FastQC, fastp, BWA, Bowtie2, HTSeq, DESeq2
PRJNA788930	35652307	Shivarathri R et al.	2022	12	NS	RNA-seq
PRJNA792028	36913408	Bing J et al.	2023	15	GCA_002759435.2, GCF_002775015.1	HiSat2, StringTie, DESeq2, BWA
PRJNA904261	37769084	Santana DJ et al.	2023	6	B8441 (Clade I)	RNA-seq
PRJNA10152963	8493178	Bing J et al.	2024	141	B8441 (GCA_002759435.2)	HiSat2, StringTie, DESeq2, BWA
PRJNA902676	38722168	Yang B et al.	2024	40	B11220, B11221 (Clades II, III)	Kallisto, DESeq2
PRJNA103603	739480072	Li J et al.	2024	22	Clade IV	RNA-seq
PRJNA108600	339455573	Wang TW et al.	2024	13	B8441 (Clade I)	HISAT2, STAR, DESeq2
PRJEB57846	39297640	Rhodes J et al.	2024	12	NS	WGS, RNA-seq
PRJNA101282	140468551	Chauhan A et al.	2025	16	B8441, B11220 (CGD)	FastQC, fastp, Bowtie2, HTSeq, DESeq2
PRJNA113916	640099908	Phan-Canh T et al.	2025	15	B8441 (GCA_002759435.2)	FastQC, fastp, cutadapt, STAR, featureCounts
PRJNA120897	540530673	Yang G et al.	2025	9	Clade I	RNA-seq

BioProject	PMID	Authors	Year	Runs	Reference Genome	RNA-seq Tools
PRJNA123283040066990		Chauhan M et al.	2025	6	Clade I	RNA-seq
PRJNA129177540863525		Vidal-Montiel A et al.	2025	6	GCA_003014415.1, GCA_034640365.1 (Clades III, IV)	FastQC, Trimmomatic, STAR, featureCounts, DESeq2

NS = Not specified in available text. Data extracted from PMC full text via EuropePMC and NCBI E-utilities APIs. 44 additional BioProjects had no linked publications.

Supplementary File 1: Galaxy workflow diagrams for re-analyses. *Source:* https://github.com/nekrut/claude-projects/tree/main/rnaseq/santana24_PRJNA904261/analysis and https://github.com/nekrut/claude-projects/tree/main/rnaseq/wang24_PRJNA1086003/analysis

Supplementary File 2: Gene mapping tables with LFC correlation values. *Source:* Same repositories as above

Supplementary File 3: Analysis reports with complete methodological details. *Source:* ANALYSIS_REPORT.md in each repository

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